Enrollment No.: _____

Darshan UNIVERSITY

Darshan Institute of Engineering & Technology
B.Tech. | Sem-4 | Summer-2025

Course Code	:	2101CS403	**Date**	:	17-04-2025
Course Name	:	Operating System	**Duration**	:	150 Minutes
			Total Marks	:	70

Instructions:
1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

Q.1 **(A)** Explain Operating System as a resource manager. **4**

(B) Explain Monolithic Kernel with the help of diagram. **3**

OR

Explain layered system with the help of diagram.

(C) Define operating system. Explain any five services/functions of operating system. **7**

OR

Define operating system. Explain any five types of operating system.

Q.2 **(A)** Implement using FCFS Scheduling algorithm. Consider Five Processes P1 to P5 arrived at same time. They have estimated running time 10, 2, 6, 8 and 4 seconds, respectively. Find the average turnaround time and average waiting time for FCFS Scheduling algorithm. **4**

(B) Explain Process states transition with the help of diagram. **3**

OR

Define Process Control Block (PCB). Explain different fields of Process Control Block (PCB).

(C) Explain FCFS algorithm with example. **7**

OR

Explain SJF algorithm with example.

Q.3 **(A)** What is Deadlock? Explain four conditions of deadlock. **4**

(B) Explain shared lock variable using algorithm. **3**

OR

Explain Disabling interrupts using algorithm.

(C) Explain Readers & Writer Problem with algorithm. **7**

OR

Explain Producer Consumer problem.

Q.4	**(A)**	What is memory? Explain different types of memory.	**4**
	(B)	Explain First Fit memory allocation algorithm with example.	**3**

OR

Explain Best Fit memory allocation algorithm with example.

	(C)	Consider the following reference string. Solve and calculate the page fault rates for FIFO page replacement algorithm. Assume the memory size is 3 page frame. 1,2,3,4,5,3,4,1,	**7**

OR

Consider the following reference string. Solve and calculate the page fault rates for LRU page replacement algorithm. Assume the memory size is 3 page frame.
1,2,3,4,5,3,4,1

Q.5	**(A)**	What is I/O devices. List different types of I/O devices.	**4**
	(B)	Explain different file operations.	**3**

OR

Explain different attributes of file.

	(C)	Explain FIFO disk scheduling Algorithms with example.	**7**

OR

Explain SSTF disk scheduling Algorithms with example.

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